

PERSPECTIVE

Risk analysis of disinformation weaponized against critical networks

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Abstract

The vulnerability of critical networks to disinformation creates significant risks of disruption with potentially severe societal consequences. Maintaining secure and resilient networks, including infrastructure and supply chain networks, is important for ensuring economic productivity along with securing the health and well-being of society. An over-the-horizon threat to critical networks deals with adversaries who attack such networks indirectly by altering the consumption behavior of unwitting users influenced by weaponized disinformation. The proliferation of disinformation through various online platforms could pose a significant and evolving challenge able to compromise the resilience of critical networks. In this perspectives article, we review the literature in this area and offer some future research directions aimed at protecting networks from weaponized disinformation, enhancing their robustness, resilience, and adaptability.

KEYWORDS

critical networks, disinformation, disruptions, infrastructure, misinformation

1 | INTRODUCTION

The fragility and vulnerability of critical networks has exposed society to risks of disruption with severe consequences. Maintaining secure and resilient networks, including infrastructure and supply chain networks, is vital for ensuring economic productivity and the health and well-being of society. Disruptions to these networks can have substantial consequences ranging from power outages, transportation disruptions, and water and sanitation issues to cybersecurity breaches (Chang, 2016). For example, in the Texas Blackout in February 2021, a winter storm caused widespread power outages across Texas, leaving millions without heat or electricity for days. Some residents resorted

to burning furniture to stay warm, whereas others shared food and water with neighbors. The outage highlighted the vulnerability of the state's power grid and the need for better preparation for extreme weather events. An example of the transportation vulnerability is the Colonial Pipeline Hack in May 2021, where a ransomware attack on the Colonial Pipeline, which supplies nearly half of the fuel consumed on the East Coast of the United States, caused widespread fuel shortages and panic buying. Likewise, supply chain disruptions resulting from natural disasters and the recent COVID-19 pandemic can lead to shortages of critical items, such as food and medical supplies (Sodhi et al., 2023; Soltanisehat et al., 2023; Spieske et al., 2022) as well as critical minerals that are necessary for national security and

economic prosperity (Biden-Harris Administration, 2022; Burton, 2022).

Beyond these notable examples, there is substantial literature on adversaries *directly* attacking networks, mostly from the network optimization perspective of interdiction (Smith & Song, 2020). However, much less attention has been devoted to preparing for an emerging, over-the-horizon threat: adversaries who attack such networks *indirectly* by altering the consumption behavior of unwitting users influenced by weaponized disinformation. The well-publicized disinformation surrounding the recent and current US presidential elections (Bovet & Makse, 2019; Wilde & Sherman, 2022) and pandemic vaccination adoption (Lazarus et al., 2022; Lee et al., 2022) are just two examples of many recent disinformation campaigns that have increased researchers' awareness that problems of mis/disinformation are amplified due to the wide availability and use of online platforms.

Accurate information is based on facts and is grounded in scientific evidence. Misinformation is defined as false or inaccurate information, whereas disinformation is similarly false and inaccurate but emerges from an intent to deceive the recipient (Carmi et al., 2020) and is the focus of this work. Due to the advances in new technologies allowing easy access to and dissemination of information, efforts to spread disinformation on online platforms have been on the rise (Vosoughi et al., 2018; Allcott et al., 2019), including the burgeoning use of artificial intelligence and deep fakes to create compelling disinformation messages (Bond, 2024; Satariano & Mozur, 2023; Vaccari & Chadwick, 2020). The effects of such disinformation are far-reaching as a recent study indicates that 86% of online global citizens tend to be at least initially fooled by fake news stories (Simpson, 2019). Further, "the public may not only have difficulty in identifying fake news but also fail to recognize their own deficiencies at doing so," as in a recent study, the individuals with inflated confidence in their ability to identify fake news were inadvertently the biggest spreaders of misinformation (Lyons et al., 2021). Importantly, disinformation can spread far and wide even when only shared by a small number of users. In a recent study, just 15% of the most habitual news sharers were responsible for spreading about 30%–40% of the fake news (Ceylan et al., 2023), demonstrating how prolific disinformation spread can be from a relatively small subset of users. The prevalence of disinformation and the propensity to spread suggest that users could easily be encouraged to engage with critical networks in ways that may lead to disruptions. Consider the following plausible scenarios (extended from actual events) that further illustrate how disinformation could adversely impact critical networks.

- **False alarms trigger chaos at airports:** An airline passenger tweets a false alert of a suspicious package in a restroom near Gate B32 at Chicago's O'Hare airport. Those who monitor such information from social media immediately send emergency signals. Personnel are dispatched to move passengers from the airport, canceling all outgoing flights, and preventing incoming flights from

landing. The alert creates a massive traffic jam on I-294, resulting from passengers leaving the airport. Drivers unable to enter the airport's departure area cause traffic jams at many primary, secondary, and tertiary roadways. A similar event occurred at London's Gatwick airport (Thomaselli, 2020).

- **Aging energy infrastructure plagued by aging technology and changing news patterns:** Hackers compromise a major US pipeline network, and the rampant spread of misinformation leads to a physical, real-world increase in gas prices. One critical example of this connection between small-scale hijackings and large-scale social effects can be seen in the recent Colonial Pipeline attack (Joyce, 2021). Similarly, in 2016, CNBC circulated a false story about Russian hackers penetrating the US electric grid. It took more than an hour for the story retraction, causing a spike in electricity prices and panic at a number of utility companies (Molina & Sundar, 2019). Such infrastructure is particularly at risk, given that many of the major infrastructure systems in the United States rely on decades-old technology.
- **Traffic patterns snarled by misinformation and algorithms:** Consider a scenario where a social media user falsely reports a major accident on a busy segment of roadway near a major urban center. Algorithms that compute shortest arrival time paths use this information and update Uber and Lyft routes. This causes congestion on the new routes, freeing up the old routes, which in turn creates another update in routes, resulting in oscillation effects and, in some cases, erratic driving behaviors. Similar events occurred globally (Tufnell, 2014; Barrett, 2020), with coordinated attacks exacerbated due to the widespread use of autonomous vehicles (DeBruhl & Tague, 2018).
- **Societal and public infrastructure disruption by social media influencer:** A social media influencer incites a riot after a large gathering at Union Square in New York City. The crowd, drawn by the influencer's potentially false promise of distributing free electronics and other high-end goods, sparks a chaotic scene, prompting a significant police response. The event, deemed "unpermitted" and "not sanctioned," rapidly grows to include thousands of young people who disrupt roadways and public transit networks, as well as engage in altercations with law enforcement and destruction of automobiles. The disruption costs New York City millions in damages and lost economic productivity, as well as dozens of arrests. A similar event occurred in the summer of 2023 (Marcius & Cramer, 2023).

These examples illustrate how disinformation can be weaponized to disrupt underlying critical networks and subsequently human lives along with economic productivity. Moreover, these examples show how disinformation not only could disrupt critical networks directly but also may increase their vulnerability to additional natural and manmade hazards. In this perspectives article, we offer a synthesis of recent literature on weaponized disinformation, focusing specifi-

cally on adverse effects on critical networks. Further, we identify several research directions for future investigation that can be fruitful for scholarship examining the effects of weaponized disinformation against critical networks.

2 | RECENT LITERATURE

Weaponized disinformation is a deliberately engineered manipulation designed to warp public perception and steer actions toward malicious goals (Fominaya, 2021; Freelon & Wells, 2020). It is a formidable tactic that assumes various guises, often blending into what might look like reliable news pieces, social media entries, and even seemingly harmless memes (Marwick & Lewis, 2017). This strategy preys on existing prejudices, stokes fears (Banas et al., 2024), and exploits societal rifts, warping the truth to stir up discord and provoke outrage (Bessarabova & Banas, 2023). Further spread of disinformation accelerates latent tensions, escalating conflicts and weaponizing them against particular groups or entities (e.g., Burnett et al., 2023). The ultimate aim is to shake the foundations of communities, hinder democratic processes (McKay & Tenove, 2021), manipulate community behavior, and serve the ambitions of adversaries who initiated the spread of the disinformation campaign.

Recent work has suggested that critical networks are susceptible to attacks driven by disinformation (Xu et al., 2021). This susceptibility is particularly relevant for power grids, given the wide variance in suppliers and the general lack of consumer understanding of power infrastructure (Schreiber et al., 2021). Demand-response management, in particular, is a mechanism that is vulnerable to adversarial manipulation, especially for electric power utilities. Power utilities are increasingly adopting demand-response programs to encourage consumers to reduce or change their electricity use during peak periods to reduce the impact of variability in electricity production due to renewable energy sources such as wind and solar (Lund et al., 2015).

As future energy generation scenarios become more prevalent, demand-response programs are expected to become even more critical (Schuitema et al., 2017). Demand-response messages are typically sent through text messages, emails, or social media posts and designed to influence human consumption behavior in real time (Jain et al., 2015). Raman et al. (2019) explored user behavior models to estimate the probability of adoption of fake demand-response messages and integrated such probability estimates with a power grid model to understand the adverse impact on a city's power grid while also considering defender actions to counter the spread of disinformation. Similarly, Raman et al. (2020) performed some simple experiments with how different messages (e.g., phrasing of the message and sender of the message) might influence power consumption behavior, which parameterized an influence model that incorporated aspects of cascading and linear threshold influence models. Based on the result of the influence model, the authors estimated how adoption might cause overloading, and thus removal of lines, in an urban power grid simulation. Tang et al. (2019) incorpo-

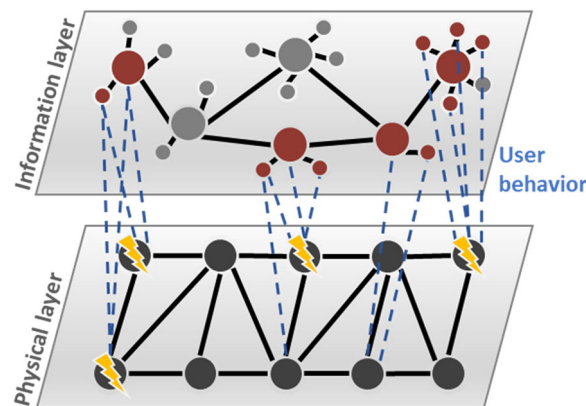


FIGURE 1 Interdependent information and physical layers proposed by Jamalzadeh et al. (2024).

rated personality traits (e.g., the “big five” (McCrae & Costa, 2004)) into an influence model and related a social network to an underlying power grid. They simulated several iterations of influence spread based on uncertain user behaviors accounting for personality traits in their model. Jamalzadeh et al. (2022) developed a hybrid approach that integrates an epidemiological model of disinformation spread (e.g., a susceptible-infected-recovered, or SIR, model) with a mixed-integer programming model to describe the performance of an underlying electric power grid. Their network optimization model examined the best protection and response actions against disinformation to minimize the disruption of the power grid.

The aforementioned research suggests that most work on the potential of disinformation to be weaponized against critical networks has been applied to the disruption of electric power grids. However, some recent research also explored impacts of disinformation on transportation networks. Waniek et al. (2021) studied the spread of false traffic alerts (e.g., accident and congestion) with the aim of manipulating the routes taken by drivers in a city through convergence (i.e., pushing drivers toward a particular location) versus divergence (i.e., pulling drivers away from a particular location) messages. The authors experimented with a simple survey of user behavior to estimate the likelihood to follow-through on a notification. They integrated such behavior estimates with a traffic simulation to measure additional trip time to quantify the adverse impact of disinformation. Khameneh et al. (2025) integrated real schedule, ridership report, and real-time alert data from the Port Authority Trans-Hudson (PATH) system in New Jersey and New York to study the impact to public transit travel time of false traffic alerts shared online. Jamalzadeh et al. (2024, 2025) conceptualized the threat of weaponized disinformation in terms of its effects on two layers, as shown in Figure 1: (i) an *information layer* being formed from social media and other social interactions and (ii) a *physical layer* representing the components of a networked cyber-physical system (e.g., infrastructure network, supply chain, and Internet of Things). Connecting these layers is the user behaviors informed by the information layer and manifested in the physical layer. As such, a weaponized disinformation attack

originates in the information layer and causes emergent human responses (e.g., changes in consumption), ultimately altering performance in the physical layer. The authors then make use of this conceptual framework to integrate an SIR model with a network optimization model to understand the impacts of disinformation on a physical multi-commodity layer. Motivated by transportation and supply chain disruptions resulting from disinformation spreading in the trucking industry (Stern, 2022; Winters, 2022), the integrated models were applied to a multi-commodity Swedish rail network. Similar to Jamalzadeh et al. (2022), the two layers were connected with a simple threshold: If a sufficient number of users in the vicinity of a physical node received and acted upon disinformation, the node was disrupted.

In the case of supply chain networks, disinformation can compromise the availability of raw material or component parts through the spread of false or misleading information (e.g., altering consumer assumptions about different characteristics of goods and services delivered by the supply chain network) by bad actors. Propagating false news about the shortage, quality, or credibility of a product could change consumption behavior and product demand (Spanaki et al., 2023). Such changes in consumption behavior were prevalent during the COVID-19 pandemic (Arafat et al., 2021; Bermes, 2021; Loxton et al., 2020). Zheng et al. (2021) developed an optimal inventory ordering policy under the threat of supply chain disruption risk (i.e., panic buying) driven by information sharing on social media platforms. Akhtar et al. (2023) explored artificial intelligence and machine learning to detect disinformation to prevent supply chain disruptions. Konstantakis et al. (2023) used an economic input–output framework to model the supply chain network impacts of disinformation shared online.

Taken together, the research discussed above represents a small yet robust literature on disinformation effects on critical networks. The research results highlight the vulnerability of networked systems to disinformation attacks. Extant models help understand the impact of disinformation on transportation and power grid performance, by developing and testing user behavior models along with hybrid approaches based on epidemiological models of disinformation spread. Research employing a conceptual framework connecting the information layer (social media and interactions) to the physical layer (infrastructure networks) may be a viable path forward for interdisciplinary work delineating how disinformation can cause disruptions through emergent human responses. The literature underscores the need for further research to understand, manage, and mitigate the threats due to disinformation in various networked systems.

3 | RESEARCH GAPS

We identify a number of research thrusts necessary for understanding the potential weaponization of disinformation against critical networks.

3.1 | Understanding social, cognitive, and affective determinants of disinformation diffusion

When creating models of disinformation diffusion, it is critical to account for social, cognitive, and affective factors underlying people's propensity to share disinformation. Although the examination of socio-scientific factors has been prolific in communication, psychology, and sociology, interdisciplinary approaches are needed to address the complexity of this issue. Accounting for social factors is key, given that online behavior such as the dissemination of disinformation is ultimately a communicative and social phenomenon. The spread of mis/disinformation is often driven by a lack of mindfulness and an inherent social desire to share information (An et al., 2023). Psychological drivers also contribute to the sharing of disinformation. Emotional states, for example, affect accuracy perceptions and decrease the ability to differentiate between fake versus actual news stories (Martel et al., 2020). Information packaged in intense emotional tones (e.g., high levels of anger) tends to spread more rapidly (Berger, 2011; Lewandowsky et al., 2012). Disinformation capitalizing on fearmongering and outrage as well as the use of intense positive affect can spread quickly, as is the case with the QAnon conspiracy theory (Bessarabova & Banas, 2023). Overall, if disinformation about power grid or supply chain networks is new or evokes a particular set of emotions, such disinformation can be quickly disseminated throughout social networks.

Similarly, understanding the motivations behind disinformation spread could help prevent dissemination of false information: For instance, people may share disinformation because they think they are helping others (Ma & Chan, 2014). Thus, it is not surprising that during COVID-19 altruistic intentions drove the spread of misinformation (Adnan et al., 2021; Apuke & Omar, 2021; Fernández-Torres et al., 2021; Malik et al., 2023). If adversaries present disinformation as important for others in the community to know, such misinformation can be spread willingly and fast through mediated and word-of-mouth communication. In addition, transdisciplinary models of disinformation diffusion can benefit from accounting for factors from the social amplification of risk framework (SARF) (Kasperson et al., 1988) proposing that “certain aspects of hazard events and their portrayal in mediated and other sources interact with psychological, social, institutional, and cultural processes in ways that might attenuate (decrease) or amplify (increase) perceptions of risk and, through this, shape behaviour” (Pidgeon & Henwood, 2010).

Besides accounting for variables from such approaches as SARF, theory-based preemptive strategies—prebunking or inoculation (e.g., Banas & Bessarabova, 2023)—may be used to make people mindful of adversarial techniques. Such strategies might remind information receivers to check messages for faulty logic and inconsistencies so common in disinformation (e.g., Banas & Miller, 2013; Banas et al., 2024;

O'Mahoney et al., 2023) or teach people how to spot a troll on social media (Lees et al., 2023) or how to recognize prevalent misinformation techniques (e.g., Roozenbeek & van der Linden, 2019, 2020), including a manipulative use of emotions (Bessarabova & Banas, 2023). Reminders to ascertain the credibility of the information source along with the use of other media literacy techniques could be helpful at preventing disinformation spread even in situations of high uncertainty. In fact, inoculation strategies can reduce uncertainty, as recent research suggests that reduced uncertainty is an important mediator of resistance against conspiracy-theory disinformation (Bessarabova & Banas, 2024). Preemptive strategies like these are particularly important, given that correcting misinformation can be challenging to accomplish once information has been disseminated into a network (Vraga et al., 2023).

Future research examining the effects of adversarial disinformation, targeting critical networks, should integrate these socio-scientific approaches with systems engineering to document the systemic influences of disinformation, accounting for social, cognitive, and affective factors in disinformation spread. Employing such interdisciplinary approaches may help better predict human behavior by connecting the information and physical layer depicted in Figure 1.

3.2 | Understanding disinformation diffusion mechanisms in the context of critical networks

Within social networks, individuals exchange information to influence others, wherein influence is an action “to induce a change in behavior of another that conforms to the influencing actor’s desires” (Hamill et al., 2007). Influence is operationalized as a new or altered behavior, resulting from direct exposure to communication by another actor. Each user communicates with (or influences) peers, and likewise, users are influenced by numerous other peers. Several diffusion models have been proposed to study the spread of such influence (Kempe et al., 2003; Perc et al., 2019), including variations on the SIR epidemiological model described previously (e.g., Jamalzadeh et al., 2022, 2024). However, recent work suggests that an epidemiological SIR model of the spread of disinformation falls short in capturing the complexity of the spread of false information. SIR-type models have limitations in the context of social media, notably failing to account for the underlying topology of social-network connections that shape the spread of information (Nekovee et al., 2007). Indeed, disinformation tends to originate in clusters, and its spread between clusters is dictated by characteristics of those clusters (e.g., age, political orientation, and education). A possible alternative is to combine mathematical models of influence spread over networks, such as independent cascades and stochastic linear threshold models (Kempe et al., 2003), with models of physical infrastructure networks. An important open challenge in such approaches, however, is the different semantics of nodes: In physical networks, nodes are associated with geospatial locations, whereas the spread of misinformation over social networks need not be

tied to spatial locality. Recent work on multiplex agent-based models (ABMs) shows that such complexity can, in fact, be modeled with newer forms of ABMs (An et al., 2023).

Algorithms and increased automation impact network diffusion both within and between the layers depicted in Figure 1. For example, filtering algorithms on social media can impact the social network by selecting which content is prioritized (Chitra & Musco, 2020). Moreover, link prediction and recommendation methods impact the connectivity structure within the social network (Santos et al., 2021). Additionally, algorithmic techniques can screen some of the content, including disinformation, thereby reducing the likelihood it propagates from a source to any followers (Cen & Shah, 2021). Finally, automation in device management, such as software-defined networks, impacts the structure of the physical layer, as well as its connectivity with the information layer (Kreutz et al., 2014).

Future research testing approaches to modeling interactions between physical and information networks would help advance knowledge in this space and would help to develop new approaches to understanding the tangible ramifications of disinformation spread. Additionally, effective modeling of the interplay between physical and information networks would help to advance theorizing about the relationship between these networks, particularly in situations where automatic detection algorithms are being deployed.

3.3 | Competing interdiction models in the information and physical layers

The adversary proposed in this work seeks to interdict the performance of the physical layer through the manipulation of user behavior through the information layer. And naturally, there is a desire by physical layer operators (e.g., utilities) to interdict the spread of disinformation that is causing such user behavior. These opposing motivations create a competition for interdiction across the two layers. Generally, network interdiction problems involve two operators, an attacker and a defender, who act sequentially to neutralize the impact of each other’s respective attack and defense efforts (Israeli & Wood, 2002; Lim & Smith, 2007). In most cases, these actions involve the disruption of network components by the attacker and protection actions by the defender. A handful of attacker–defender models dealt with the spread of influence, including influence blocking (He et al., 2012; Jia et al., 2020), manipulating opinion diffusion (Bredereck & Elkind, 2017), and controlling elections (Wilder & Vorobeychik, 2018, 2019). In the problem described in this work, the attacker deliberately spreads disinformation, whereas the defender reacts afterward to neutralize the disinformation spread. Defender decisions include where to place resources (e.g., designing incentives, adding new intentional messaging crafted to establish new norms and nudge against disinformation (Xiao, 2022)) to spread factual information and limit the spread of disinformation. Although models for protection and interdiction exist for interdependent networks (e.g., the

tri-level optimization problems of Ghorbani-Renani et al. (2020, 2021)), an area of future research deals with representing the competing actions across layers, especially when there is a lone adversary but different defenders governing the different layers (e.g., social media application vs. electric power utility).

3.4 | Long-term impacts to power generation portfolios

Highlighting a particular critical network—the power grid—the prevalence of disinformation targeting the renewable energy transition presents a considerable barrier to sustainability agendas (Simon, 2022a, 2022b). Recent opposition to the renewable energy transition espouses health, economic, and nuisance effects that are often unfounded (Simon, 2022b). The misperceptions about renewable energy are especially common in rural communities, where solar and wind infrastructure would likely be placed and where many users engage with social media platforms that provide easy access to disinformation (Borch et al., 2020; Fergen et al., 2021). For example, some media outlets blamed solar and wind energy sources for the widespread power outages during the extreme cold temperatures experienced in Texas in February 2021 (Swenson & Lajka, 2021; Gearino, 2022). In reality, renewable energy sources outperformed forecasts during 90% of the blackout, whereas natural gas plants underperformed, relative to wind energy, due to frozen pipes and wellheads (Lovins & Ramana, 2021; FERC, 2021). Other myths about the unreliability of a grid powered by renewables, the necessity of continued reliance on fossil fuels, and the importance of retaining nonrenewable energy to ensure variability in energy profiles have similarly been debunked (Lovins & Ramana, 2021). The propagation of such disinformation coincides with long-standing efforts to discredit and distort evidence of climate change (Marsh, 2022; Supran et al., 2023), as well as a recent “stark comeback for climate denial,” especially on social media (ISD, 2023). As such, there is a need to fully study the long-term impacts of disinformation on the future of energy portfolios, including understanding attitudes and beliefs regarding renewable energy, determining how to inoculate against such disinformation, and developing energy transition constraints and decarbonization scenarios to feed into national energy planning models. In particular, it is important to understand the economic, environmental, and social impacts of energy transition scenarios on different socially vulnerable communities as driven by disinformation.

3.5 | Disproportionate effects of disinformation on vulnerable communities

Considering that different subgroups of society may be more susceptible to false information, such sectors would have enhanced risks of disproportionate direct effects of

disinformation (Seo et al., 2021). These adverse outcomes may be particularly critical for vulnerable communities due to: (i) limited access to resources, (ii) greater language barriers, or (iii) limitations in literacy/numeracy, among others. Moreover, considering that vulnerable communities and underrepresented minorities are often more susceptible to natural and anthropogenic hazards and disaster events, their already larger exposure to diverse risks associated with direct damages caused by an attack to their infrastructure may be further exacerbated by disinformation affecting their planning, speed, and overall effectiveness of preparedness, mitigation, and recovery strategies. Therefore, it is important to understand how different communities may be affected disproportionately, directly and indirectly, by disinformation. Moreover, it is critical to develop tailored strategies for disinformation protection that consider unique sociocultural characteristics of different communities, capitalizing on their own individual strengths, while accounting for any potential vulnerabilities.

It is important to understand and quantify how diverse decisions to protect critical networks from weaponized disinformation may affect different communities and develop fairness-based strategies to help promote network performance equitably. To this end, future work should consider how resources are distributed, for example, among networks serving different locations or vulnerable groups (fairness in input) as well as the socio-economic impacts of the different policies (fairness in output). Even though the strategies to allocate resources to mitigate disinformation may seem to have an average positive impact across the population, their effects are often distributed inequitably throughout different communities. Therefore, such strategies should account for an equitable allocation among communities such that vulnerable communities are especially protected. There have been diverse fairness-based allocation schemes proposed in the literature and included in optimization models, including (i) social welfare, (ii) Kalai Smorodinsky, (iii) max–min, (iv) proportional fairness, and (v) entropy, among others (Sampat & Zavala, 2019; Moshebah et al., 2024). Social welfare allocation schemes seek to maximize the aggregate utility over all individuals. Kalai Smorodinsky allocation implies that all individuals have a proportional potential gain, each receiving the same fraction of allocation with respect to their maximum achievable utilities. In a max–min allocation, individuals that receive the lowest allocations are prioritized. In a scheme seeking proportional fairness, the aggregate proportional changes associated with each individual cannot be enhanced with respect to other possible allocations. In an allocation seeking to reduce entropy, the goal is to minimize variations from the egalitarian allocation (where everyone is assigned the same). As noted, different fairness-based schemes have different considerations in mind; thus when ensuring optimization models represent fairness, it is imperative to procure a multidimensional perspective that accounts for different needs and constraints of all individuals as well as the communities' they form.

4 | CONCLUDING REMARKS

The proliferation of disinformation through various online platforms could pose a significant and evolving challenge to achieving resilient critical networks. The amplification of false information creates a profound impact on decision-making, counteraction, and response.

The current methods for managing risks associated with critical networks are generally effective at addressing more tangible and predictable risks. However, these methods are often inadequate when it comes to dealing with over-the-horizon risks posed by the spread of weaponized disinformation. These events have severe consequences but are typically dismissed as outliers or deemed impossible to predict using conventional risk management approaches.

In this perspectives article, we discuss a few potential areas of exploration for new theory, methodology, and application to address the weaponization of disinformation against critical networks. These areas include the study of socio-scientific drivers of disinformation spread, the relationship between disinformation spread and its impact on critical networks, protection–interdiction models across information and physical layers, the role disinformation plays in future network portfolio decisions, and uneven nature of disinformation and network impacts to different communities.

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